Phase-III
Solution Architecture

Date	05 March 2026
Team ID	Team 154
Project Name	Insurance Fraud Detection using ML
Maximum Marks	4 Marks

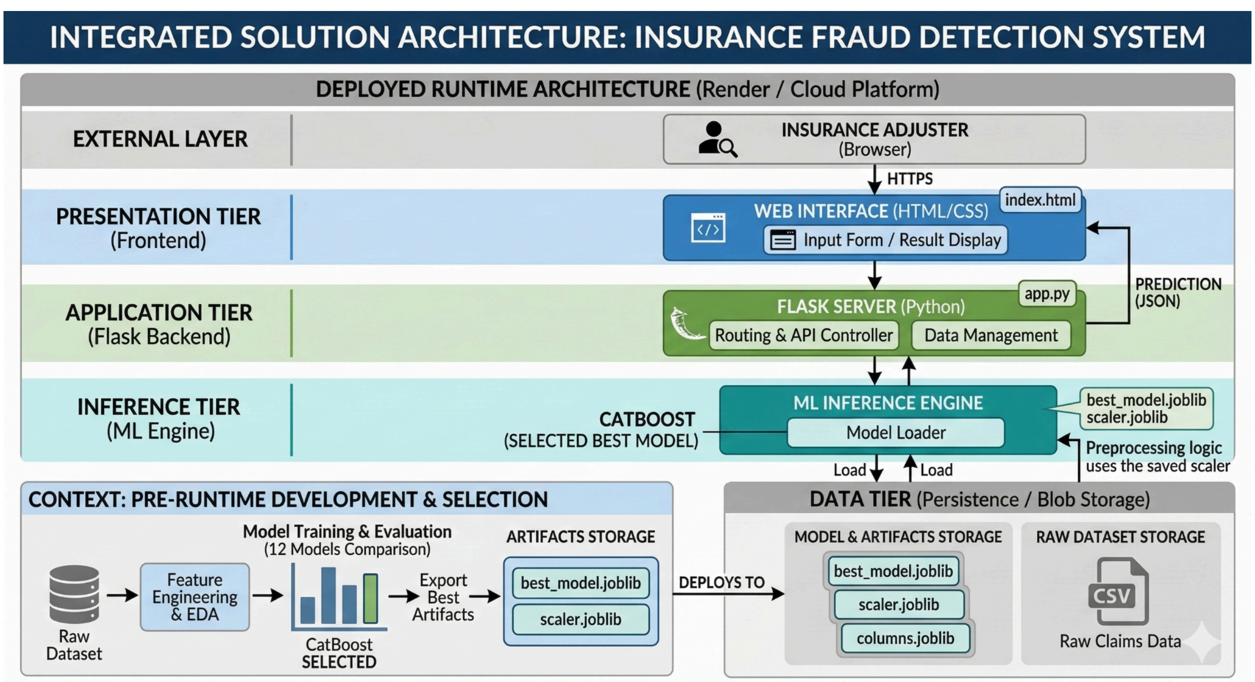
Architecture Overview:

Input Layer (Frontend): A web-based interface (HTML/CSS) where insurance investigators enter claim details such as incident type, policy info, and total claim amount.

Service Layer (Flask Backend): Acts as the "orchestrator." It receives the data from the frontend, applies the necessary scaling using the saved scaler.joblib, and passes it to the ML model.

Inference Layer (ML Engine): The heart of the system where the best-performing model (selected from 12 candidates) processes the input to determine the risk score.

Database/Storage: Local storage containing the original dataset and serialized model files (.joblib) required for runtime.



Development Phases

Following industry standards, the solution is defined and delivered through these key phases:

Phase	Description
Data Engineering	**Handling missing values, cleaning '?' entries, and feature scaling.**
Model Research	**Training and comparing 12 algorithms (Random Forest, CatBoost, etc.).**
System Integration	**Connecting the Flask backend with the serialized ML models.**
UI Development	**Building the web form for real-time investigator interaction.**